AI-102 · Microsoft Azure AI Engineer

学习笔记 PDF 报告

生成日期：2026-07-29 · 数据截止：2026-07-29 · 内容来源：Microsoft Learn + 个人研究

执行摘要

本 PDF 是 chasejwang/AI102 GitHub 仓库的完整内容汇编。13 个 markdown 笔记，~120 KB，覆盖 Microsoft 认证 AI-102 考试的全部 6 个功能组 + 23 个子技能 + 10 个官方文档。

> ★ 考试 AI-102 已于 2026-06-30 退休。但 Microsoft Foundry 仍是当前生产栈，6 个功能组（Foundry 规划 / GenAI / Agentic / Vision / NLP / 知识挖掘）仍是工程师的 competency model。本笔记适合 portfolio + 求职 + 微软生态 展示。
>
> ★ 关键发现：(1) LUIS 已退休 2025-10-01，改用 CLU 或 Foundry Agent Service；(2) Cognitive Services → AI Services → Foundry 三大重命名；(3) Agentic solution 是新增 5-10% sub-skill。

仓库结构

类型	文件数	内容
入口	2	README.md + .gitignore
顶层笔记	3	Study Guide + Index + Master Mind Map
Reference Docs	10	10 个 Microsoft Learn 官方文档
总计	15	约 120 KB · 全部 200 OK 在线

Part 1 · 主笔记 (00 - AI-102 Study Guide.md)

AI-102 · Microsoft Azure AI Engineer Associate

> ** Exam AI-102 was retired on June 30, 2026** at 11:59 PM Central Standard Time. This study guide is preserved for **historical reference, portfolio, and job-market currency** — many job postings still reference "AI-102" or "Azure AI Engineer" as a target credential.
>
> Source: <https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/ai-102>

Exam at a glance

Audience profile

> As a Microsoft Azure AI engineer, you build, manage, and deploy AI solutions that leverage Azure AI.

Responsibilities (full lifecycle):

- Requirements definition and design
- Development
- Deployment
- Integration
- Maintenance
- Performance tuning
- Monitoring

Tools:

- Python + C# (primary)
- REST APIs + SDKs
- Image processing, video processing, NLP, knowledge mining, generative AI

Soft skills: Translate architect's vision, collaborate with data scientists / data engineers / IoT specialists / infrastructure admins.

Skills measured (6 functional groups)

```
pie title AI-102 Weight Distribution "Plan & Manage" : 22 "Generative AI" : 17 "Agentic" : 7 "Computer Vision" : 12 "NLP" : 17 "Knowledge Mining" : 17
```

> **★ Most heavily weighted**: Plan/Manage (22%) + NLP (17%) + Knowledge Mining (17%) = 56% of exam.

> **★ New since last version**: "Implement an agentic solution" (5-10%) — reflects Microsoft Foundry Agent Service GA in 2025.

Sub-skills by functional group

1. Plan and manage an Azure AI solution (20-25%)
2. Implement generative AI solutions (15-20%)
3. Implement an agentic solution (5-10%)
4. Implement computer vision solutions (10-15%)
5. Implement natural language processing solutions (15-20%)
6. Implement knowledge mining and information extraction (15-20%)

Reference documentation (10 crawled notes)

Study resources (from official study guide)

Critical rebrand: "Cognitive Services" → "Foundry"

> All 10 reference doc URLs from the original study guide **redirect to new locations** as of 2026. The product family has been renamed **three times**:

```
Cognitive Services (2017) ↓ Azure AI Services (2023) ↓ Microsoft Foundry / Foundry Tools (2025) ↓ Foundry Models (separate brand for hosted LLMs, formerly Azure OpenAI Service)
```

> **★ Exam tip**: On the AI-102 exam, you will see all three brand names interchangeably. Know they refer to the same product family.

Change log (Dec 23, 2025)

> **★ Bottom line**: The functional groups are stable, but the product branding was overhauled. New "agentic" sub-skill added.

Exam-day reality (retired exam)

> **If you took AI-102 before 2026-06-30**: the certification is valid for **1 year from pass date**, then you must renew via free online assessment.

>

> **If you didn't take it**: the credential is no longer available. Microsoft's recommended successor is **Azure AI Foundry Engineer** (replacement cert not yet announced as of 2026-07-29 — watch the certification page).

For portfolio + LinkedIn, the AI-102 still demonstrates:

- Knowledge of the Microsoft Foundry ecosystem (the current production stack)
- Skills in 6 functional areas: Foundry planning, GenAI, Agentic, Vision, NLP, Knowledge Mining
- Familiarity with Azure-specific branding and SDK patterns

Related (across your vault)

- `../Vision/00 - Vision` _(external vault link) — your own Career vision doc
- `Profile` _(external vault link) — your background
- `Data Agent project` _(external vault link)
- `NLP-Learning` _(external vault link)
- `Resume` _(external vault link) — for showcasing this cert
- `Job Search 2026` _(external vault link)

_{Study guide content crawled 2026-07-29 · All 10 official reference docs saved with current 2026 names (Microsoft Foundry / Foundry Tools) and historical rebrand context. Source: <https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/ai-102>}

_{Source: Obsidian Vault snapshot from 2026-07-29. Part of the AI-102 study pack. Exam AI-102 was retired 2026-06-30 — these notes are preserved for portfolio + Microsoft Foundry competency reference.}

> ★ 注：Module-Mind-Map.md 的 Mermaid master mind map 在 PDF 中无法渲染。请直接查看 GitHub 仓库渲染。

Part 2 · 10 个 Reference Doc 摘要

下面是从 10 个 Microsoft Learn 官方文档爬取整理的笔记。每篇含：能力、场景、SKU、API 端点、连接服务、考试要点。

Microsoft Foundry 伞形

Azure AI Services / Microsoft Foundry

> Microsoft Foundry (formerly "Azure AI Services", formerly "Cognitive Services") is the umbrella platform for building, deploying, and governing AI applications and agents at scale on Azure. It bundles pre-built AI capabilities (Foundry Tools), foundation models from OpenAI and partners (Foundry Models), an agent orchestration runtime (Foundry Agent Service), and a control plane for governance — all accessible via a unified portal (ai.azure.com), REST APIs, and SDKs in Python / C# / JS / Java.

Key Capabilities

• **Foundry Tools** — pre-built AI services for Vision, Speech, Language, Translator, Document Intelligence, Content Understanding, Face, Immersive Reader (each is a billable Azure resource with its own endpoint/key).

• **Foundry Models** — catalog of hosted models sold by Azure (OpenAI GPT, DeepSeek-R1, Mistral, Llama, Phi, etc.) with pay-as-you-go or provisioned deployment types.

• **Foundry Agent Service** — runtime to orchestrate, host, and share AI agents; integrates prompt agents, hosted agents, and Foundry IQ for knowledge grounding.

• **Foundry IQ** — knowledge-base grounding layer that connects enterprise data (Azure AI Search, Cosmos, Blob) to agents via retrieval-augmented generation.

• **Foundry Control Plane** — centralized governance for token limits, agent lifecycle management, observability/tracing, evaluation, and Content Safety enforcement.

• **Observability & Evaluation** — tracing for agent calls, dashboards for gen-AI apps, batch evaluation of agentic workflows, AI Red Teaming Agent.

• **Foundry Local** — runs open-source LLMs (HuggingFace, etc.) on-device / on-edge for free inference.

Common Use Cases

• Customer-service copilots grounded on internal documentation (Foundry Agents + IQ).

• Image and document analysis pipelines in enterprise workflows (Vision + Document Intelligence).

• Multilingual speech-to-speech translation for contact centers (Speech + Translator).

• Content moderation on user-generated content (Content Safety).

• RAG applications combining OpenAI models with Azure AI Search indexes.

• Edge/on-prem LLM inference for low-latency or air-gapped scenarios (Foundry Local on Azure Local / Arc).

Service Tiers / SKUs (if applicable)

• **Standard (S0)** — pay-per-transaction, the default tier for nearly all Foundry Tools.

• **Free (F0)** — limited free quota for Vision / Language / Speech / Translator (used in quickstarts).

• **Provisioned / PTU** — reserved throughput for OpenAI / Foundry Models (predictable cost, low latency).

• **Connected / Disconnected containers** — Docker containers for Vision / Speech / Language / Translator for on-prem deployments.

Key APIs / SDK Methods (if shown)

SDKs: Python, C#, JavaScript/TypeScript, Java. Foundry also offers an MCP Server and a VS Code extension.

Connections to Other Services

• **Azure AI Search** — knowledge store for RAG / Foundry IQ.

• **Azure Blob Storage / Cosmos DB / OneLake** — grounding data sources.

• **Azure Machine Learning** — model training, pipelines, AutoML.

• **Content Safety** — scanning prompts/responses before/after model calls.

• **Azure Arc / Azure Local** — edge deployment of Video Indexer and Foundry Local.

• **Microsoft Entra ID (AAD)** — authentication for all resource endpoints (key-based or Microsoft Entra token).

Exam-Relevant Notes

• **Naming history** — Microsoft rebranded the stack twice: **Cognitive Services** (2017) → **Azure AI Services** (2023) → **Microsoft Foundry / Foundry Tools** (2025). On the retired AI-102 exam, expect to see all three names in case studies; know they refer to the same product family.

• The Foundry umbrella separates **Models** (you bring/host the LLM) from **Tools** (pre-built AI capabilities you call as APIs).

• All Foundry Tools resources use **multi-region endpoint pairs** and require a **region selection** when provisioned (pricing varies by region; check `azure.microsoft.com/global-infrastructure/services/?products=cognitive-services`).

• Authentication options: **API Key** (resource page) or **Microsoft Entra ID** (preferred for production / RBAC).

• Responsible AI: Foundry ships **Content Safety** and an **AI Red Teaming Agent** for safety evaluation.

Azure AI Vision

Azure AI Vision

> Azure AI Vision (formerly "Computer Vision", a Foundry Tool) provides cloud-based algorithms that analyze image content. Submit an image (raw bytes or URL) and the service returns tags, captions, detected objects, OCR text, people/landmarks, and (via the Spatial Analysis add-on) real-time video analytics. Common in retail, media, accessibility, and document-digitization workloads.

Key Capabilities

• **Image Analysis 4.0** — single-call returns captions, dense captions, tags, objects, people, and OCR; choose features via the `features` parameter.

• **Optical Character Recognition (OCR)** — extract printed/handwritten text; supports 164+ languages; returns text lines + bounding-box polygons.

• **Image tagging** — returns a confidence-rated list of content tags (e.g. "indoor", "dog", "car").

• **Image captioning** — generates a natural-language sentence describing the image (neutral / accessible styles).

• **Object detection** — returns bounding boxes + class labels for detected objects.

• **Spatial Analysis** (video) — detects people/objects in live video streams; runs as an edge container.

• **Face detection** (limited) — basic attributes; full Face API lives at `/ai-services/face/` (separate, access-restricted resource).

Common Use Cases

• **Product cataloging** — auto-tag and caption e-commerce product photos.

• **Document digitization** — OCR scanned forms into searchable text.

• **Accessibility** — generate alt-text for screen readers.

• **Content moderation** — flag adult/racy content (use **Content Safety** for that instead — Vision returns only "adult/gory" booleans).

• **Retail analytics** — Spatial Analysis to count people in-store zones (requires Azure IoT Edge).

• **Vehicle / landmark recognition** — domain-specific models return make/model or famous landmarks.

Service Tiers / SKUs (if applicable)

OCR Read v3.x and Image Analysis 4.0 are billed per image / per feature.

Key APIs / SDK Methods (if shown)

SDKs: Python, C#, Java, Node.js.

Connections to Other Services

• **Content Safety** — preferred over Vision's `adult` flag for moderation.

• **Azure AI Search** — index extracted OCR text for full-text search.

• **Form Recognizer / Document Intelligence** — better than raw OCR for structured forms/invoices.

• **Storage Blob** — source of batch images (URL-based calls).

• **Custom Vision** — sibling service for training your own image classifier (now part of Vision Studio / Foundry).

Exam-Relevant Notes

• **Naming history** — **Computer Vision** (Cognitive Services) → **Azure AI Vision** (AI Services) → **Vision (Foundry Tool)**. The REST endpoint namespace stayed `vision/v3.2` for backward compatibility.

• OCR Read API handles **printed + handwritten** text; returns **text + bounding polygon** + style info (handwritten/other).

• Image Analysis 4.0 returns **confidence scores (0–1)** and **bounding boxes** — use these for filtering.

• Spatial Analysis requires **Azure IoT Edge / Arc**; not a cloud-only service.

• **Face API is a SEPARATE resource** with limited access (Microsoft requires an intake form for facial recognition / identification / liveness).

• Service supports both **API Key** and **Microsoft Entra ID** authentication.

Azure AI Video Indexer

Azure AI Video Indexer

> Azure AI Video Indexer (AVI) is a cloud **and** edge service that extracts rich insights from video and audio — transcripts, translation, sentiment, OCR-on-screen, face detection (limited), speaker identification, scene/shot segmentation, labels, and named entities. Two deployment options: a fully managed cloud SaaS, or **Video Indexer enabled by Azure Arc** for hybrid/edge/on-prem live analysis.

Key Capabilities

• **Speech transcription** — multi-speaker, multi-language; supports 50+ languages with auto language ID.

• **Translation** — translate the transcript into 50+ languages.

• **Sentiment analysis** — per-utterance positive/neutral/negative.

• **OCR** — text appearing on screen is extracted with timestamps.

• **Face detection / identification** — detects faces and can match against a custom **Person Model** (limited-access feature).

• **Speaker identification** — clusters who spoke when; named against a custom **Speaker Identification** model.

• **Scene / Shot detection** — visual scene changes and keyframe thumbnails.

• **Labels & Named Entities** — visual content tags, people, locations, organizations.

• **Audio effects** — identifies music, applause, silence, etc.

• **Live & near-real-time analysis** — via Arc-enabled edge deployment.

Common Use Cases

• **Media & broadcast** — auto-generate archive metadata for searchable content libraries.

• **Enterprise compliance** — screen recorded meetings/calls for policy violations.

• **Sports & highlight reels** — auto-detect plays, score changes, crowd reactions.

• **Accessibility** — generate captions/subtitles and chapter markers.

• **Law-enforcement forensics** — evidence indexing (subject to Limited Access policy).

• **Real-time safety** — Arc edge deployment for detecting PPE / safety gear / equipment on factory floors.

Service Tiers / SKUs (if applicable)

Key APIs / SDK Methods (if shown)

• **Upload** — `POST {endpoint}/Accounts/{accountId}/Videos?name=...&videoUrl=...` (or multipart upload).

• **Get insights** — `GET /Accounts/{accountId}/Videos/{videoId}/Index` → JSON with `videos[].insights.transcript`, `faces`, `topics`, etc.

• **Get thumbnail** — `GET /Accounts/{accountId}/Videos/{videoId}/Thumbnails/{thumbnailId}`.

• **Embed widget** — iframe player that surfaces insights in a web UI.

• SDKs available; most integrations use the **REST API** directly. Widgets ship as **JS / React** components.

Connections to Other Services

• **Azure Media Services / Azure Blob Storage** — source video ingestion.

• **Azure Arc / Azure Kubernetes Service** — required for edge deployment.

• **Event Grid** — fires events on `JobStateChanged`, `InsightsFinished`.

• **Azure AI Search** — index AVI insights for natural-language search ("find the meeting where Bob said X").

• **Foundry Agent Service** — agent can call AVI as a tool to ground answers in video archives.

Exam-Relevant Notes

• AVI is a **separate resource type** (not the same as Azure Media Services); you create an `Azure AI Video Indexer` account from the Azure Marketplace (it provisions a managed identity + storage + AMS account behind the scenes).

• **Two deployment modes** are an exam favorite: (1) **Cloud SaaS**, (2) **Arc-enabled** for edge.

• **Limited Access features** = restricted behind Microsoft's Responsible AI review (face ID, celebrity ID, emotion).

• AVI returns a **JSON insights document** with arrays for transcript lines, faces, topics, labels — exam questions often ask which field to read (e.g. `videos[0].insights.transcript[].text`).

• For **real-time / live streaming analysis** use the Arc deployment, not the cloud SaaS.

• **Person Model** (custom face identification) must be trained and assigned to the account before face identification works.

Azure AI Language (含 CLU / Q&A; / LUIS 退休)

Azure AI Language (and LUIS)

> Azure AI **Language** is the current Foundry Tool for NLP: sentiment, key phrases, entities, language detection, summarization, translation, and **Conversational Language Understanding (CLU)** — Microsoft's replacement for the now-deprecated **LUIS**. LUIS still works (Classic portal at `luis.ai`) but Microsoft retired it on **October 1, 2025**; new builds should use **CLU** (part of Language Service) or **Azure OpenAI / Foundry Agent Service** for intent-style tasks.

Key Capabilities

• **Sentiment Analysis & Opinion Mining** — per-sentence and per-aspect sentiment + targets.

• **Key Phrase Extraction** — pulls salient noun phrases.

• **Named Entity Recognition (NER)** — general + PII / redaction.

• **Language Detection** — auto-identify 125+ languages.

• **Summarization** — extractive and abstractive summaries of documents/conversations.

• **Question Answering** — extract answers from a knowledge base / unstructured docs.

• **Conversational Language Understanding (CLU)** — successor to LUIS; intents, entities, utterances, model training, deployment.

• **Custom Text Classification** — multi-label single-label classifiers you train on your own data.

• **Custom Named Entity Recognition (NER)** — train custom entity extractors.

• **Text Analytics for Health** — clinical/medical NER & relations.

Common Use Cases

• **Customer-feedback analytics** — sentiment + key-phrase dashboards.

• **PII redaction** in transcripts and chat logs (Health & Financial entities).

• **Conversational AI** — CLU replaces LUIS for bot/agent intent parsing ("Book a flight to Tokyo").

• **Knowledge-base Q&A** — power chatbots from FAQs / docs.

• **Document summarization** — auto-summarize long emails / articles.

• **Content classification** — auto-route support tickets by topic.

Service Tiers / SKUs (if applicable)

Key APIs / SDK Methods (if shown)

SDKs: Python (`azure-ai-textanalytics`, `azure-ai-language-questionanswering`, `azure-ai-language-conversations`), C#, JS, Java.

Connections to Other Services

• **Bot Framework Composer / Bot Service** — historically used LUIS; now uses CLU or Foundry Agent Service.

• **Foundry Agent Service** — modern alternative; agents handle intent via foundation models instead of CLU.

• **Azure AI Search** — index extracted entities / summaries for RAG.

• **Speech Service** — pipeline Speech→Text→Language for voice bots.

• **Azure Blob Storage / Cosmos DB** — source data for Q&A / custom NER.

Exam-Relevant Notes

• **Naming history** — **LUIS** (2017) → **Language Service / CLU** (2023) → **Language (Foundry Tool)** (2025). LUIS portal lives at `luis.ai`; documentation now at `previous-versions/azure/ai-services/luis/`.

• **LUIS retired Oct 1 2025.** On the (retired) AI-102 exam, expect questions on both LUIS **and** CLU — know that CLU uses **utterances, intents, entities** identical to LUIS, but training is done via `ConversationAuthoringClient` against an **Azure AI Language resource** (not a LUIS resource).

• LUIS concepts to remember: **Utterances** (training examples), **Intents** (the action), **Entities** (data extracted), **Phrase Lists** (boost features), **Prebuilt Domains**, **Batch Testing**.

• CLU adds **orchestration** — a single project can route among multiple apps (QnA, CLU, LUIS).

• **Endpoint keys** vs **Authoring keys**: authoring requires a separate key/role assignment.

• Knowledge of when to use **CLU vs Foundry Agent** is increasingly test-relevant — agents are preferred for new LLM-style conversational systems; CLU for deterministic, low-latency intent classification.

Azure AI Speech

> Azure Speech (now branded **Azure Speech in Foundry Tools**) provides speech-to-text, text-to-speech, real-time speech translation, and live AI voice conversations as a fully managed cloud service, also available in containers for edge / on-prem deployment. Used by Microsoft internally for Teams captions, Office 365 dictation, and Edge Read Aloud, and by external customers for call centers, language learning, captioning, audiobooks, and conversational voice agents.

Key Capabilities

• **Speech to text** — real-time streaming transcription, fast transcription of pre-recorded audio, and batch transcription for large volumes; supports custom acoustic / language / pronunciation models for noisy or domain-specific audio.

• **Text to speech** — neural voices with SSML control over pitch, rate, volume, pronunciation; standard voice gallery plus private **custom neural voice** branding.

• **Text-to-speech avatar** — synthesizes a photorealistic digital video of a person speaking the input text, sync or async.

• **Speech translation** — real-time speech-to-speech and speech-to-text translation across many languages.

• **Language identification** — detect which language(s) are spoken in audio; usable standalone or chained with STT / translation.

• **Pronunciation assessment** — scores accuracy and fluency for language-learning apps.

• **Voice Live (formerly part of gpt-realtime / voice agents)** — low-latency speech-to-speech voice agent SDK for conversational AI experiences.

• **LLM speech (preview)** — LLM-enhanced STT / translation with better context understanding and prompt-tuning ("transcribe" and "translate" tasks).

Common Use Cases

• Live captioning for meetings, streams, and accessibility (Teams-class captions).

• Call center transcription + sentiment / PII redaction for analytics.

• Audiobook / e-learning narration with neural voices.

• In-car navigation and voice assistants.

• Language learning apps with pronunciation scoring.

• Conversational voice agents / bots (Voice Live) for customer support.

• Synthetic video avatars for content creation or training.

Service Tiers / SKUs

Pricing is per-feature (STT, TTS, translation each have their own meter). Standard tiers are pay-as-you-go; committed-use tiers exist for high volume. Container SKUs are available for on-prem deployment for compliance. See the Speech pricing page for current rates. Notable SKU axes:

Key APIs / SDK Methods

• **Speech SDK** — primary SDK in C#, C++, Go, Java, JavaScript, Objective-C/Swift, Python. Exposes `SpeechRecognizer`, `SpeechSynthesizer`, `TranslationRecognizer`, `VoiceLiveClient`.

• **Voice Live SDK** — newer low-latency SDK in C# / Python (Java/JS preview) for speech-to-speech agents.

• **Speech Transcription SDK** — Java / Python for high-quality transcription apps.

• **Speech CLI (`spx`)** — code-free command-line wrapper over most SDK features.

• **REST APIs** — speech-to-text, batch transcription, text-to-speech, custom voice, batch synthesis, batch avatar.

• **Speech Studio** — no-code UI for building projects, then reference assets via SDK / REST.

Connections to Other Services

• **Microsoft Foundry** — Speech is exposed as a **Foundry Tool**; surface via the **Speech MCP tool** to Foundry agents.

• **Azure OpenAI / Foundry Models** — multimodal speech inputs (GPT-4o audio) and LLM speech features.

• **Multimodal content** — ingestion into Azure AI Search for multimodal search.

• **Bot Service / Direct Line** — classic integration path for speech-enabled bots.

• **Sovereign clouds** — Azure Government, Azure operated by 21Vianet (China).

• **Containers** — run on Kubernetes, Azure Container Instances, or on-prem for disconnected scenarios.

Exam-Relevant Notes

• **Naming history**: Cognitive Services Speech → Azure Speech → **Azure Speech in Foundry Tools** (current). The exam may still expect "Cognitive Services Speech" or "Azure Speech"; both refer to the same service.

• Custom Speech = acoustic + language + pronunciation training; needs Speech project in Speech Studio.

• Custom Neural Voice is **gated** — requires application and Microsoft's responsible-AI sign-off (limited access).

• **Three STT modes** matter for the exam: real-time (streaming), fast (pre-recorded, sync), batch (async, large). Know which to pick for a given scenario.

• Pronunciation Assessment is a discrete feature with its own SDK method (`PronunciationAssessmentConfig`) — language-learning scenario.

• Speech Translation returns interim + final results; uses `TranslationRecognizer`.

• Speech-to-text REST API current stable version is **2025-10-15** (migration guide published).

• Audio Content Creation = neural TTS for audiobooks / in-car nav; uses SSML.

• Containers are the answer when "data must stay on-prem" / "compliance" / "disconnected" scenarios come up.

• Speech is part of the **Foundry multi-service resource** — can share an endpoint with other Foundry Tools.

Visual

```
flowchart LR Mic[Microphone / Audio File] --> SDK[Speech SDK / CLI / REST] SDK -->
STT[Speech to Text] SDK --> TTS[Text to Speech] SDK --> Trans[Speech Translation]
SDK --> LID[Language ID] SDK --> PA[Pronunciation Assessment] SDK --> VL[Voice
Live] STT --> Apps[Your App / Bot] TTS --> Apps Trans --> Apps VL -->
Agents[Foundry Agents] STT -.custom model.-> CS[Custom Speech] TTS -.custom
model.-> CV[Custom Neural Voice] SDK -.container.-> Edge[On-prem / Edge]
```

Azure AI Search

> Azure AI Search (formerly **Azure Cognitive Search**, before that **Azure Search**) is a fully managed cloud search service that unifies full-text, vector, hybrid, and multimodal retrieval over your enterprise data. It powers both traditional index-based search and modern retrieval-augmented generation (RAG) via **agentic retrieval**, and underpins **Foundry IQ** — the knowledge layer for Microsoft Foundry agents.

Key Capabilities

• **Two retrieval engines**:

• **Classic search** — single-request, index-first, low-latency (no LLM in the loop).

• **Agentic retrieval** — multi-query, LLM-assisted planning + subquery decomposition + parallel retrieval + semantic reranking + merged response.

• **Query types** — full-text (Lucene-style inverted index), vector (similarity over embeddings), hybrid (RRF fusion of text + vector), multimodal (text + images in one pipeline), fuzzy, autocomplete, geo-spatial, filters.

• **Indexing** — JSON documents only; supports push (direct upload) or pull (indexer from a data source).

• **AI enrichment (skillsets)** — chunking, embedding generation, OCR, entity extraction, custom skills; the foundation of *cognitive search*.

• **Integrated vectorization** — built-in embedding + chunking at index or query time.

• **Remote knowledge sources** — query external stores (e.g. SharePoint, web) live without re-indexing.

• **Enterprise security** — Microsoft Entra auth, Azure Private Link, document-level access control, role-based access, security filters.

• **Relevance tuning** — scoring profiles, semantic ranker, synonyms, faceted navigation.

Common Use Cases

• Enterprise search apps (intranet, document libraries).

• RAG pattern for chatbots / agents grounded on private data.

• E-commerce product search with filters and faceting.

• Knowledge mining pipelines that ingest blobs, extract entities, and serve an index.

• Multimodal search across PDFs containing text + images.

• Permission-aware agentic RAG over SharePoint, OneLake, Blob Storage, Cosmos DB.

Service Tiers / SKUs

Two pricing models are now offered:

Within Dedicated, classic tiers from low to high: Free, Basic, S1, S2, S3, plus higher-capacity tiers. Replicas (read scale) and partitions (write scale / storage) are configurable independently. Storage is fixed per partition.

> **Exam legacy**: AI-102 was written against the older S1/S2/S3/Storage-Optimized tier names. The concepts still map — replicas = query throughput, partitions = index storage and write throughput.

Key APIs / SDK Methods

• **REST API** — `https://<service>.search.windows.net/` with versioned API versions (e.g. `2024-05-01-preview`).

• **Azure SDKs** — `@azure/search-documents` for Python, JavaScript, Java, .NET.

• **Key REST surfaces**:

• `Indexes` — create/update/delete index (schema with fields, suggesters, scoring profiles, semantic config).

• `Indexers` — pull data from sources, run skillsets, write to index.

• `Skillsets` — cognitive skill pipeline (OCR, split, embed, custom Web API skill).

• `Data Sources** — connection info for indexers.

• `Documents` — upload/merge/delete/lookup (`@search.action`, `search=`, `vector=`, `select=`, `filter=`).

• `Knowledge Bases` + `Knowledge Sources` — for agentic retrieval.

• `Suggesters` / `Autocomplete`.

• **Search Explorer** — built-in panel in Azure portal for ad-hoc queries.

Connections to Other Services

• **Microsoft Foundry (Foundry IQ)** — Search is the managed knowledge layer for Foundry agents; knowledge bases attach to agents.

• **Azure OpenAI / Foundry Models** — provides grounding content for LLMs; embeddings model integration for vector indexing.

• **Data sources** — Azure Blob Storage, Azure Data Lake Storage Gen2, Azure SQL, Cosmos DB, SharePoint, OneLake, and web URLs.

• **Azure AI Foundry SDKs** — full programmatic access for orchestration.

• **Logic Apps** — can drive indexers / data ingestion workflows.

• **Azure Machine Learning** — custom skills deployed as web services can run inside skillsets.

Exam-Relevant Notes

• **Naming history**: Azure Search → Azure Cognitive Search → **Azure AI Search** (current). The exam will accept both "Cognitive Search" and "AI Search" answers.

• **Classic search** vs **agentic retrieval** — know the difference:

• Classic: one query, one index, no LLM in the loop, returned ranked documents.

• Agentic: knowledge base → LLM plans → multiple subqueries → parallel retrieval → semantic reranking → synthesized answer with activity log + references.

• **Skillsets** = the "cognitive" piece: OCR, split, entity recognition, key-phrase extraction, language detection, custom Web API skill, plus embedding generation for vectors.

• **Indexer pattern** — schedule-driven, change-tracking-aware pull from supported data sources; output to an index.

• **Integrated vectorization** builds embedding into the indexer pipeline so you don't have to embed offline.

• **Hybrid search** uses Reciprocal Rank Fusion (RRF) to combine BM25 text + vector results.

• **Semantic ranker** is a separate, billable add-on that re-ranks top results using a deeper model.

• **Security trimming** — `search.in` filter on

... [本节在 PDF 中截断，完整内容见 GitHub 仓库] ...

Azure OpenAI (Foundry Models pivot)

Azure OpenAI Service

> Azure OpenAI Service provides REST API access to OpenAI's GPT, Codex, embedding, image, audio, and reasoning models hosted on Azure infrastructure. It is now a subset of **Microsoft Foundry Models** ("Foundry Models sold by Azure"), unified under the Microsoft Foundry portal at `ai.azure.com`. The legacy `azure/ai-services/openai/` docs URL redirects to `/azure/foundry/`.

URL REDIRECT — read this first

The original study-guide URL `https://learn.microsoft.com/en-us/azure/ai-services/openai/` redirects to the Microsoft Foundry hub at `https://learn.microsoft.com/en-us/azure/foundry/`. The Azure OpenAI-specific content now lives under the Foundry Models catalog at `/azure/foundry/foundry-models/concepts/models-sold-directly-by-azure` (with the `?pivots=azure-openai` pivot). The data, control plane, and feature set are unchanged — only the documentation surface was consolidated.

Key Capabilities

• **Chat / text generation** — GPT-5.x series, GPT-4.1, GPT-4o, GPT-4 Turbo; supports system prompts, structured outputs, function/tool calling, and parallel tool calls.

• **Reasoning models (o-series)** — `o1`, `o3`, `o4-mini`, `codex-mini` for advanced problem solving; use the Responses API for the latest reasoning features.

• **Embeddings** — `text-embedding-3-large`, `text-embedding-3-small`, `text-embedding-ada-002` for vector representations used by AI Search, RAG, similarity.

• **Image generation** — DALL·E 3 and later image models, called via the Images API.

• **Video generation** — `sora-2` (Foundry Models sold by Azure).

• **Audio** — GPT-4o audio models with speech-in / speech-out, low-latency conversational interactions, and audio generation.

• **Fine-tuning** — supervised fine-tuning and (for some models) RLHF / DPO on your own data.

• **Content filtering / Responsible AI** — built-in Content Safety integration (prompt shields, jailbreak detection, harmful content categories).

• **Provisions and quota** — per-region, per-model deployment types (Standard, Global Standard, Provisioned, Reserved, Batch, etc.).

Common Use Cases

• Chatbots, copilots, and RAG over enterprise data (usually paired with Azure AI Search).

• Code generation / explanation (Codex, GPT-5.x-codex).

• Document summarization, extraction, classification.

• Embeddings for semantic search, clustering, recommendations.

• Image generation for marketing, design, content.

• Speech-to-speech voice agents via GPT-4o audio or Voice Live.

• Automated reasoning / planning with o-series models.

• Function-calling agents that invoke external tools / APIs.

Service Tiers / SKUs (Deployment Types)

Azure OpenAI uses deployments of a specific model. Each deployment has a deployment type that governs pricing & throughput:

> **Exam legacy**: AI-102 was written against the older model names (GPT-3.5, GPT-4, GPT-4 Turbo, embeddings-ada-002, DALL·E 3). The newer GPT-5.x, o-series, gpt-oss, and Sora families are Foundry-era additions; you still need to know the *patterns* (chat completions, embeddings, DALL·E, fine-tuning, content filtering).

Key APIs / SDK Methods

• **Chat Completions API** (`/chat/completions`) — primary text-generation endpoint. Supports `messages`, `tools`, `response_format`, `temperature`, `max_tokens`.

• **Responses API** — newer unified endpoint combining chat completions + tool calls + state; preferred for reasoning models and agentic flows.

• **Embeddings API** (`/embeddings`) — vector representations.

• **Images API** (`/images/generations`, `/images/edits`) — DALL·E.

• **Video API** — Sora.

• **Audio API** — speech-to-text, text-to-speech, audio generation (realtime + batch).

• **Fine-tuning API** — `POST /fine_tuning/jobs` for supervised fine-tuning.

• **Assistants API** (now superseded by Foundry Agent Service) — thread/run, code interpreter, file search, function calling.

• **Azure OpenAI SDKs** — `openai` Python library pointed at Azure endpoint, plus `@azure/openai` / `Azure.AI.OpenAI` (C# / Java / JS).

• **Foundry SDK** — unified SDK for all Foundry Models including OpenAI family.

Connections to Other Services

• **Microsoft Foundry** — primary home; Azure OpenAI is now a Foundry Model `kind`.

• **Foundry Agent Service** — wrap OpenAI models in hosted agents with tool calls, threads, knowledge bases.

• **Azure AI Search** — typical RAG pairing: Search retrieves, OpenAI generates.

• **Content Safety** — every chat-completions call passes through Content Safety filters by default.

• **Azure Blob Storage / OneLake** — feed fine-tuning datasets and Assistants file_search.

• **Azure AI Services multi-service resource** — historically could share an endpoint; modern recommendation is a dedicated Foundry resource.

• **Logic Apps / Functions / APIM** — wrap OpenAI calls with governance, caching, and token budgets.

Exam-Relevant Notes

• **Naming history**: Azure OpenAI Service → Azure OpenAI in Foundry → **Foundry Models sold by Azure** (current). Older docs calls it "Azure OpenAI"; the exam will still use that name.

• **Resource type** in ARM: `Microsoft.CognitiveServices/accounts` with `kind=OpenAI` (or Foundry equivalent). Deployment is a child resource.

• **Provisioning** — OpenAI access historically required a form / application; that gate is now relaxed for most accounts but high-rate-limit tiers may still need approval.

• **Authentication** — Microsoft Entra (recommended) **or** API key. Always prefer Entra in production.

• **Content filters** — built-in, configurable (input + output filters per category). Prompt Shields detect jailbreak / indirect prompt injection.

• **Token limits** — every model has a **context window** (input + output) and a separate **max output tokens**.

... [本节在 PDF 中截断，完整内容见 GitHub 仓库] ...

Azure AI Document Intelligence

Azure Document Intelligence in Foundry Tools

> Azure Document Intelligence is a cloud-based Microsoft Foundry tool (formerly Azure AI Document Intelligence, formerly Form Recognizer) that uses machine-learning models to automate data extraction from documents in applications and workflows. It is essential for enhancing data-driven strategies and enriching document search capabilities, and is consumed by developers building document automation into apps or by knowledge-mining pipelines that need structured output from unstructured PDFs/images.

Key Capabilities

• **Prebuilt models** for common document types: Read (OCR + handwriting), Layout (structure, tables, figures), Financial Services & Legal, US Tax, and Personal Identification.

• **Custom extraction models** — train on your own labeled forms to extract domain-specific key-value pairs and fields.

• **Custom classification models** — train a classifier to route documents to the right extraction model or workflow.

• **Query field extraction** — extract specific fields on demand at inference time, without retraining the model.

• **Add-on capabilities** — optional features (e.g., high-resolution, formula, font property extraction) layered onto prebuilt or custom models.

• **Batch document analysis** — submit many documents in one request for async processing.

• **Retrieval-Augmented Generation (RAG)** support — clean structured output that drops cleanly into vector indexes / Foundry vector search.

• **Confidence and accuracy scores** returned with every prediction, for downstream gating.

• **Multiple deployment options** — cloud endpoint, or Docker **container** for disconnected/edge use.

Common Use Cases

• Automated invoice / receipt processing (AP automation).

• Contract and legal-document clause extraction.

• Identity document verification (passports, driver's licenses, US tax forms).

• Mortgage / loan application form processing.

• Knowledge mining over enterprise document archives (PDF library → search index).

• RAG preprocessing — converting unstructured PDFs into clean chunks + metadata for vector search.

Service Tiers / SKUs

Pricing is per-page for both prebuilt and custom models, with separate metering for add-on capabilities. Document Intelligence is provisioned as a single-resource service (one endpoint per resource).

Key APIs / SDK Methods

• `POST /documentModels/{modelId}:analyze` — start an analysis job (returns `Operation-Location`).

• `GET /documentModels/{modelId}:analyze/{resultId}` — poll for results.

• **Studio** at `documentintelligence.ai.azure.com` for no-code labeling, training, and testing.

• SDKs: **Python**, **.NET (C#)**, **JavaScript/TypeScript**, **Java**.

• Auth: API key, Microsoft Entra ID (managed identity supported for storage access).

• Input formats: PDF, JPEG/JPG, PNG, BMP, TIFF, HEIF; Office files via Office conversion.

Connections to Other Services

• **Azure Blob Storage** — typical source of documents (read via SAS token or managed identity).

• **Azure AI Search / Foundry vector search** — RAG downstream of extraction.

• **Azure OpenAI / Foundry Models** — used in RAG pipelines over extracted content.

• **Logic Apps / Functions / Data Factory** — orchestration and ETL of document batches.

• **Microsoft Entra ID** — authentication and managed-identity access to storage.

Exam-Relevant Notes

• **Name evolution** (definitely on legacy exam material): **Form Recognizer** → **Azure AI Document Intelligence** → **Azure Document Intelligence in Foundry Tools** (current branding). All three refer to the same service.

• It is a **Foundry Tool** (the new umbrella for what used to be "Azure AI Services" / "Cognitive Services"). The branding is `Cognitive Services → Azure AI Services → Foundry Tools`.

• Know the difference between **prebuilt** (Read, Layout, prebuilt invoice/receipt/ID) and **custom** (extraction & classification) models — exam scenarios will pick one.

• **Layout** model is the workhorse: extracts text, tables, selection marks, figures, and structure — useful for RAG preprocessing and any "give me structured JSON from this PDF" scenario.

• **Read** model = OCR + handwriting (the "next-generation" OCR; replaces older Recognize Text).

• Custom models require a labeled training set in the Studio; compose models let you group multiple custom models behind one endpoint.

• For AI-102: recognize the four core Azure AI solution areas (vision, speech, language, decision) and where Document Intelligence fits (knowledge mining + information extraction).

• **Version note** (current docs, 2026): the **v3.0 API retires March 30, 2029** — exam-scenarios may reference v3.0 → v4.0 migration.

Visual

```
flowchart LR A[PDF / Image<br/>in Blob Storage] --> B[Document Intelligence<br/>Endpoint] B --> C{Model} C -->|prebuilt| D[Read / Layout /<br/>Invoice / Receipt / ID] C -->|custom| E[Custom Extraction<br/>or Classifier] C -->|add-on| F[HighRes / Formula /<br/>Query Fields] D & E & F --> G[Structured JSON<br/>+ confidence scores] G --> H[Downstream:<br/>AI Search / Foundry<br/>Logic Apps / DB]
```

1 小时 9 单元训练模块

Plan and Prepare to Develop AI Solutions on Azure

> This is the **first module in the AI-102 learning path** ("Develop generative AI apps in Azure"). It is a 1-hour, 9-unit beginner module that gives you the mental model for the whole exam: what AI capabilities Azure offers, what Microsoft Foundry and Foundry Tools are, which SDKs you use, and what responsible-AI rules apply. If you only read one piece of Microsoft Learn before the exam, read this one.

Key Capabilities (Learning Objectives)

By the end of this module you should be able to:

• Identify common AI capabilities you can implement in applications

• Describe **Microsoft Foundry** and considerations for using it

• Describe **Foundry Tools** and considerations for using them

• Identify appropriate developer tools and SDKs for an AI project

• Describe considerations for **responsible AI**

Module Structure (9 units, ~1 hour)

Reward: 1000 XP, generic badge. Module authored with assistance from AI (per the disclosure at the bottom of the page).

The Big Idea (what an AI-102 candidate must internalize)

AI capability categories

The module frames Azure AI in terms of common capabilities you'll bolt into an app:

• **Machine learning** (custom models, Azure ML)

• **Anomaly detection** (Univariate / Multivariate in Foundry Tools)

• **Computer vision** (Image Analysis, Custom Vision, Face, Document Intelligence)

• **Natural language processing** (Language service: sentiment, NER, translation, CLU/LUIS)

• **Speech** (Speech to Text, Text to Speech, Speaker Recognition, Translation)

• **Knowledge mining + information extraction** (Document Intelligence, AI Search)

• **Generative AI** (Foundry / Azure OpenAI, prompt flow, RAG)

Microsoft Foundry (the hub)

• Unified **portal** for building, evaluating, deploying generative-AI and traditional AI solutions.

• Hosts **Foundry projects** where you assemble models, data, indexes, and evaluation flows.

• Replaces / subsumes older "AI Studio" and "Azure ML Studio" experiences.

Foundry Tools (the API services)

• The new umbrella brand for what was **Azure AI Services**, which was itself the rename of **Cognitive Services**.

• Discrete REST + SDK services you call from your app: Vision, Language, Speech, Document Intelligence, Content Safety, Translator, etc.

• Each tool is provisioned as its own resource (or grouped under a multi-service resource) and accessed via an endpoint + key (or Microsoft Entra ID).

Developer tools & SDKs

• **Languages**: Python and C# are the two the cert page calls out by name; the exam assumes comfort in one of them.

• **SDKs**: per-service SDKs (e.g. `azure-ai-documentintelligence`, `azure-ai-vision-imageanalysis`, `azure-ai-inference`) plus the older `azure-cognitiveservices-*` legacy SDKs.

• **REST**: every Foundry Tool has a REST API — you should be comfortable reading a `curl` snippet and the `Ocp-Apim-Subscription-Key` / `Authorization: Bearer` headers.

• **Tooling**: VS Code + AI Foundry extension, Azure CLI (`az`), VS / Visual Studio, GitHub Codespaces.

Responsible AI

• Six Microsoft principles: **fairness, reliability & safety, privacy & security, inclusiveness, transparency, accountability**.

• Practical implications on the exam: choosing the right Foundry Tool for a sensitive use case, content filtering via **Azure AI Content Safety**, transparency notes, data residency.

Connections to Other Services

• **Microsoft Foundry** ← hosts / orchestrates Foundry Tools and Foundry model deployments.

• **Foundry Tools** ← consumed from your app via SDK or REST.

• **Azure OpenAI** ← typically surfaced inside Foundry projects; not a "Foundry Tool" per se (it's a model-deployment service), but adjacent.

• **Azure AI Search** ← common companion for RAG patterns over extracted documents.

• **Microsoft Entra ID** ← auth for the SDKs and for managed-identity access to data sources.

Exam-Relevant Notes

• **Branding chain** (high-yield trivia): **Cognitive Services → Azure AI Services → Foundry Tools**. The "AI services" hub is now the **Foundry** portal; individual services are **Foundry Tools**.

• For AI-102 you should be able to **map a business requirement to the right tool**:

• "extract fields from invoices" → Document Intelligence

• "summarize call-center audio" → Speech-to-Text + Language

• "build a chat assistant over our docs" → Foundry + Azure OpenAI + AI Search (RAG)

• "moderate user-generated content" → AI Content Safety

• "detect objects in product photos" → Image Analysis / Custom Vision

• Know the difference between **Foundry** (the platform/portal) and **Foundry Tools** (the individual AI services) — exam scenarios will mix the terms.

• Auth: API key vs Microsoft Entra ID vs **managed identity** (preferred for apps accessing storage / AI Search).

• This module is part of the **"Develop generative AI apps in Azure"** learning path — which lines up with the AI-102 domain *Implement generative AI solutions*.

Visual

认证主页 + 退休通知

Microsoft Certified: Azure AI Engineer Associate (AI-102)

> This is the **official certification page** for the Azure AI Engineer Associate. It defines the role, the assessed skills, and exam logistics. **Important for your study plan: the page now carries a retirement banner** — the certification and its renewal assessment are retired. You're studying this for historical / portfolio / job-market currency, not for a fresh cert. Target audience: AI engineers who build and ship AI solutions on Azure using Foundry Tools, AI Search, and Azure OpenAI.

Retirement Status

• The cert page displays: **"This certification and the renewal assessment are retired."**

• Background: AI-102 was retired **June 30, 2026**. The skill set is being absorbed into newer role-based credentials and Microsoft Applied Skills badges.

• Studying AI-102 still makes sense for: job-market resume keywords, interviews that reference it, internal upskilling, and as a foundation for the replacement credentials.

Role Definition (from the cert page)

> "As a Microsoft Azure AI engineer, you build, manage, and deploy AI solutions that leverage Azure AI."

Responsibilities across the full lifecycle:

• Requirements definition and design

• Development

• Deployment

• Integration

• Maintenance

• Performance tuning

• Monitoring

You collaborate with solution architects (translating their vision) and data scientists, data engineers, IoT specialists, infrastructure admins, and other developers to build complete end-to-end AI solutions and integrate AI into other applications.

Required language skills

• **Python** and **C#** are the two explicitly called out.

• Must be comfortable with **REST APIs and SDKs**.

Required Azure portfolio knowledge

• Components of the Azure AI portfolio (Foundry, Foundry Tools, Azure OpenAI, AI Search).

• Available **data storage options** (Blob, Data Lake, Cosmos DB, etc.).

• **Responsible AI principles** applied in practice.

What the Exam Assesses (the 6 domains)

(Domain weights shift per exam update — verify against the most recent skills measured PDF before sitting the exam. The six domain headings above are verbatim from the cert page.)

Exam Logistics

• **Duration:** 100 minutes

• **Format:** Proctored, includes interactive (lab) components

• **Languages:** English, Japanese, Chinese (Simplified), Korean, German, French, Spanish, Portuguese (Brazil), Chinese (Traditional), Italian

• **Retake policy:** 24 hours after 1st attempt; longer waits for subsequent attempts.

• **Accommodations** available.

Prerequisites / Adjacent Certs

• **Azure Fundamentals (AZ-900)** is the typical recommended (not required) starting point.

• **Azure AI Fundamentals (AI-900)** is a useful warm-up — overlaps in Foundry Tools concepts but is less implementation-focused.

• Common next step after AI-102: **Azure Solutions Architect Expert (AZ-305)**, **Azure Developer Associate (AZ-204)**, or **Microsoft Applied Skills** badges in targeted areas.

Exam-Relevant Notes

• This is the canonical **"what does an Azure AI Engineer do"** document — read it once at the start of your study plan and again before the exam.

• The six domains map cleanly to the modules in the AI-102 learning path. Treat each domain as a study unit.

• "**Implement an agentic solution**" is the newest domain — reflects the shift toward Foundry-based agent / copilot patterns. For legacy study material, this is the area to supplement with the latest Foundry agent docs.

• Branding chain to memorize: **Cognitive Services → Azure AI Services → Foundry Tools / Microsoft Foundry** — the cert page itself still references "Azure AI services" and "Azure AI Search" in the certification tagline.

• Even though the cert is retired, the **skills measured** document is still the single best study outline you can get for free.

• The page advertises a **Practice Assessment** (`assessmentId=61`) and an **Exam Sandbox** (look-and-feel demo) — both still valuable for last-mile prep.

Visual

```
flowchart TD A[Azure AI Engineer] --> B[Plan & Manage] A --> C[Implement GenAI] A --> D[Implement Agents] A --> E[Computer Vision] A --> F[NLP] A --> G[Knowledge Mining /<br/>Information Extraction] B --> H[Foundry + Foundry Tools<br/>+ Entra ID + Cost] C --> I[Azure OpenAI +<br/>Foundry + RAG] D --> J[Foundry Agent Service<br/>+ Tool Use] E --> K[Image Analysis /<br/>Custom Vision / Face /<br/>Document Intelligence Vision] F --> L[Language / CLU /<br/>Speech / Translator] G --> M[AI Search +<br/>Document Intelligence +<br/>Indexers + Skillsets]
```

总结 · 三个 Takeaway

1. 考试退休 ≠ 知识退休。AI-102 在 2026-06-30 退休，但 6 个功能组（Foundry 规划 / GenAI / Agentic / Vision / NLP / 知识挖掘）是 Microsoft Foundry 工程师的 competency model，仍是求职展示的核心。

2. 品牌重命名是高频陷阱。Cognitive Services (2017) → Azure AI Services (2023) → Microsoft Foundry (2025) + Foundry Models。三个名字指同一套产品，面试和读文档时必须知道。

3. Agentic 是新热点。Domain #3「Implement an agentic solution」(5-10%) 是 AI-102 新增 sub-skill，对应 Microsoft Foundry Agent Service + Microsoft Agent Framework——跟 LangGraph / AutoGen 等开源项目同主题。

跟现有笔记的关系

★ Career/Profile — 把 Microsoft Foundry 6 功能组加进技能矩阵

★ Career/Data-Agent — Data Agent 调研对应 AI-102 的 Agentic sub-skill

★ Career/NLP-Learning — 6 个月 NLP 学习路径覆盖 AI-102 NLP 17%

★ Career/Resume-Output — Resume 加 Microsoft Foundry 关键词

★ Career/Job-Search-2026 — 求职面试中演示 Foundry 能力

来源与连接

• GitHub 仓库：https://github.com/chasejwang/AI102

• 官方 Study Guide：https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/ai-102

• Microsoft Foundry：https://learn.microsoft.com/en-us/azure/foundry/

• 同类认证参考：DP-600 (Microsoft Fabric) — 已发布到 chasejwang/dp-600-fabric-notes 仓库